Software Engineering Lab 7

LAB 7| Fuzzing for Input Validation Bugs

Rakshan R | SRN: PES1UG24CS365

SS1_Original_Application

```
(.venv) PS C:\Acad\Sem 5\SE Lab\Lab7\L6_L7_53_bike_share_rental\LAB7> python .\bike_share_rental_parser.py
==== Running parse_ride_duration ====
Enter a ride duration as HH:MM (e.g. 01:15): 01:15
Duration in minutes: 75

==== Running format_station_code ====
Enter a station code (e.g. a12): a12
Formatted station code: A-12

==== Running parse_ride_log ====
Enter a ride log line as type,minutes,subscriber (e.g. electric,45,yes): electric,45,yes
Parsed ride log: ('electric', 45, True)
```

SS2_Failing_Hypothesis_Tests

```
=================== short test summary info ====================
FAILED test_bike_share_rental_parser.py::test_parse_ride_duration_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exc...
FAILED test_bike_share_rental_parser.py::test_format_station_code_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exc...
FAILED test_bike_share_rental_parser.py::test_parse_ride_log_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exc...
==================== 3 failed in 4.90s =====================
(.venv) PS C:\Acad\Sem 5\SE Lab\Lab7\L6_L7_53_bike_share_rental\LAB7>
```

SS3_Fixed_Application

```
(.venv) PS C:\Acad\Sem 5\SE Lab\Lab7\L6_L7_53_bike_share_rental\LAB7> python .\bike_share_rental_parser.py
==== Running parse_ride_duration ====
Enter a ride duration as HH:MM (e.g. 01:15): abc
Duration in minutes: None

==== Running format_station_code ====
Enter a station code (e.g. a12):
Formatted station code: None

==== Running parse_ride_log ====
Enter a ride log line as type,minutes,subscriber (e.g. electric,45,yes): electric,abc
Parsed ride log: None
(.venv) PS C:\Acad\Sem 5\SE Lab\Lab7\L6_L7_53_bike_share_rental\LAB7>
```

SS4_Passing_Hypothesis_Tests

```
==================== test session starts ====================
platform win32 -- Python 3.13.0, pytest-9.2.0, pluggy-1.7.0 -- C:\Acad\Sem 5\SE Lab\Lab7\L6_L7_53_bike_share_rental\LAB7\.venv\Scripts\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Acad\Sem 5\SE Lab\Lab7\L6_L7_53_bike_share_rental\LAB7
plugins: hypothesis-6.170.15
collected 3 items

test_bike_share_rental_parser.py::test_parse_ride_duration_no_crash PASSED [ 33%]
test_bike_share_rental_parser.py::test_format_station_code_no_crash PASSED [ 66%]
test_bike_share_rental_parser.py::test_parse_ride_log_no_crash PASSED [100%]

==================== 3 passed in 1.16s =====================
(.venv) PS C:\Acad\Sem 5\SE Lab\Lab7\L6_L7_53_bike_share_rental\LAB7>
```